Tayama Kirati

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github.com/Tayama-Kirati Portfolio

EDUCATION

Tribhuwan University - Thapathali Campus
Bachelor of Engineering in Computer Engineering *2024 - 2026*

Tribhuwan University - Purwanchal Campus
Bachelor of Engineering in Computer Engineering *2022 - 2024*

St. Xaviers College, Maitighar
+2 Science *2019 - 2021*

TECHNICAL SKILLS

Programming Languages: C/C++, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, Express, NodeJS, NumPy, Pandas, PyTorch, scikit-learn

Databases: SQLite, PostgreSQL, MongoDB

Tools & Platforms: Hadoop, Figma, Unity, Blender

PROJECTS

Text-to-ASL Conversion System (Minor Project) **github.com/bhattaraisushma/Snapsigns**
3D Sign Language Translation Framework *Python, NLP, T5, Blender*

- Designed and developed a text-to-ASL translation system to convert written text into American Sign Language animations.
- Utilized the T5 transformer model for natural language understanding and sentence to sign mapping.
- Created 3D animated ASL gestures using Blender for accurate and expressive sign visualization.
- Built an end-to-end pipeline integrating NLP processing with real-time 3D animation rendering.

Monocular SLAM and NeRF Fusion for 3D Reconstruction (Major Project)
Real-Time Neural Rendering and Localization *Python, OpenCV, SLAM, NeRF*

- Built a geometric SLAM pipeline generating optimized poses and sparse point clouds from monocular video.
- Integrated Depth Anything V2 to produce detailed depth maps, for accurate 3D reconstruction.
- Established a data pipeline that bridges SLAM and NeRF by providing optimized poses and dense depth maps for neural rendering.
- Training a real-time NeRF to synthesize photorealistic, dynamic 3D scenes from the SLAM and depth outputs.

MetMoMo **github.com/Tayama-Kirati/MetMoMo**
Full-Stack Food Ordering Web Application *React, Node.js, Express, MongoDB, Cloudinary*

- Built a full-stack food ordering platform with a decoupled frontend and backend architecture.
- Implemented JWT-based authentication and secure user sessions for customers and restaurant owners.
- Integrated Cloudinary for product and restaurant image uploads via Multer, storing URLs in MongoDB.
- Designed RESTful APIs with Express and Node.js to handle orders, menus, and user management.

E-Commerce **github.com/Tayama-Kirati/E-Commerce**
Modern E-Commerce Web Application *Next.js, TypeScript, Prisma*

- Developed a full-featured e-commerce website using Next.js with TypeScript for type-safe development.
- Used Prisma ORM for database schema management and efficient data querying.
- Leveraged Next.js App Router for server-side rendering and optimized page load performance.

My Space
github.com/Tayama-Kirati/My-Space

Personal Digital Content Management App — *JavaScript, Node.js, CSS*

- Built a personal content management site featuring a daily journal, book tracker, photo memory gallery, and focus sessions.
- Designed an interactive journal with date-based entry editing, monthly cover photos, and a year overview with 12 mini month cards.
- Implemented a book logging system with star ratings, personal reviews, and a stats dashboard showing reading progress.
- Created a masonry photo gallery with drag-and-drop upload, captions, mood tags, and a monthly timeline sidebar.

Rock Paper Scissors AI
github.com/Tayama-Kirati/RockPaperScissors

Adaptive AI Game using Markov Chains — *Python*

- Built a terminal-based game where an AI learns and predicts the player's move patterns using a Markov Chain transition table.
- Implemented persistent AI memory using JSON so the model retains player history across sessions.
- Designed modular OOP architecture with separate classes for AI brain, game logic, and statistics tracking.

EXPERIENCE

Fellowship, All In Foundation (AIF)
Feb 2025 – Sep 2025

- Produced a documentary on the Dom community, highlighting their cultural identity, social challenges, and lived experiences. Conducted field research and interviews to document underrepresented voices, ensuring authentic and respectful representation through storytelling